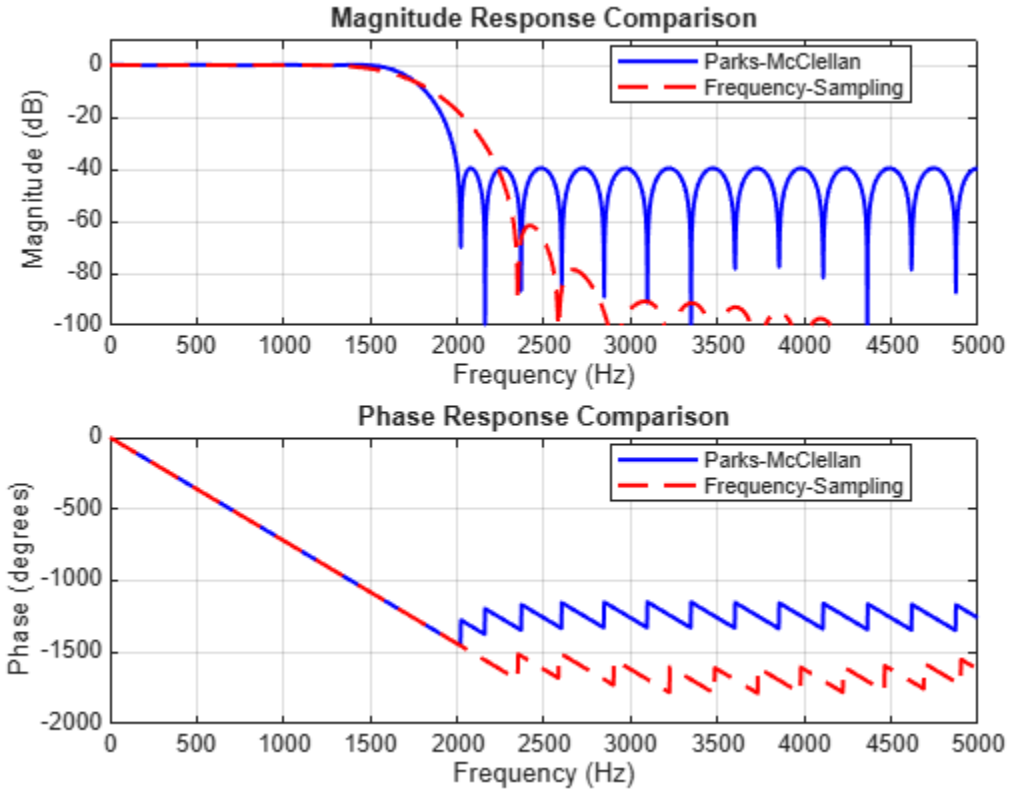
```
clc;
clear;
close all;

% EXPERIMENT 5: PARKS-MCCLELLAN vs FREQUENCY-SAMPLING

% Design parameters
Fs = 10000;       % Sampling frequency (Hz)
fp = 1500;        % Passband edge (Hz)
fs = 2000;        % Stopband edge (Hz)
N = 40;           % Filter order

% Normalized frequency values
F = [0 fp fs Fs/2] / (Fs/2);
A = [1 1 0 0];

% 1. Parks-McClellan filter design
b_pm = firpm(N, F, A);

% 2. Frequency-sampling filter design
b_fs = fir2(N, F, A);

% Display results
fprintf('EXPERIMENT 5: FIR FILTER DESIGN COMPARISON\n');
fprintf('-----------------------------------------\n');
fprintf('Sampling Frequency = %d Hz\n', Fs);
fprintf('Passband Edge      = %d Hz\n', fp);
fprintf('Stopband Edge      = %d Hz\n', fs);
fprintf('Filter Order       = %d\n\n', N);

fprintf('Parks-McClellan coefficients:\n');
disp(b_pm(1:10));

fprintf('Frequency-Sampling coefficients:\n');
disp(b_fs(1:10));

% Calculate frequency responses
[H_pm, f] = freqz(b_pm, 1, 2048, Fs);
[H_fs, ~] = freqz(b_fs, 1, 2048, Fs);

% Magnitude in dB
mag_pm = 20*log10(max(abs(H_pm), 1e-10));
mag_fs = 20*log10(max(abs(H_fs), 1e-10));

% Plot magnitude and phase responses
figure('Name', 'FIR Filter Comparison');

subplot(2,1,1);
plot(f, mag_pm, 'b', 'LineWidth', 1.5);
hold on;
plot(f, mag_fs, 'r--', 'LineWidth', 1.5);
grid on;
```

```
xlim([0 Fs/2]);
ylim([-100 10]);
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Magnitude Response Comparison');
legend('Parks-McClellan', 'Frequency-Sampling', ...
    'Location', 'best');

subplot(2,1,2);
plot(f, unwrap(angle(H_pm))*180/pi, ...
    'b', 'LineWidth', 1.5);
hold on;
plot(f, unwrap(angle(H_fs))*180/pi, ...
    'r--', 'LineWidth', 1.5);
grid on;
xlim([0 Fs/2]);
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
title('Phase Response Comparison');
legend('Parks-McClellan', 'Frequency-Sampling', ...
    'Location', 'best');

fprintf('Filter design completed successfully.\n');
```

```
EXPERIMENT 5: FIR FILTER DESIGN COMPARISON
----------------------------------------
Sampling Frequency = 10000 Hz
Passband Edge      = 1500 Hz
Stopband Edge      = 2000 Hz
Filter Order       = 40

Parks-McClellan coefficients:
  Columns 1 through 7

   -0.0015    0.0072    0.0053   -0.0010   -0.0081   -0.0072    0.0037

  Columns 8 through 10

    0.0142    0.0100   -0.0090

Frequency-Sampling coefficients:
  Columns 1 through 7

    0.0000    0.0001    0.0002   -0.0001   -0.0008   -0.0009    0.0008

  Columns 8 through 10

    0.0035    0.0029   -0.0039

Filter design completed successfully.
```



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